

Criterion	3PH IBR fault	SLG IBR fault	Advantage
$I_{\min}$ blocking	$I_a = k_{\text{ibr}}$ Blocked if $k < 0.10$ (Region A, 29% envelope)	$I_a = I_1 + I_2 + I_0$ Never blocked (I_0 from network)	SLG: no Region A
Distance relay Z_m	$Z_m = \alpha Z_{1L}$ (k.i.b.r. cancels) Correct measurement	$Z_m \gg \alpha Z_{1L}$ ($I_0 \gg I_1$: ratio broken) Z_m over-estimated	3PH better Z_m accuracy
Distance relay zones	Zone 1 always ($\alpha \leq 0.80, k \geq 0.10$) Zone 2/3 fail (3PH mem)	Zone 3 only ($\alpha \lesssim 0.35$) Zone 1/2: never reached	3PH: better zone coverage
Polarisation source	Memory only ($T_{\text{mem}} = 100 \text{ ms}$) Zone 2/3 fail	Permanent (healthy phases) All zones viable	SLG: better pol. source
87L chain coverage	100% selective (21/21 TR-27) $t \approx 20 \text{ ms}$	100% selective (via 87LN0, TDCS) $t \approx 20 \text{ ms}$	Both: 87L primary
Net result	87L + Z1 (Z2/3 via POTT TR-34)	87L + Z3* (*close-in only)	87L sole fast detector for both types